How do neural nets learn?

Serkan Sulun

serkan.sulun@inesctec.pt

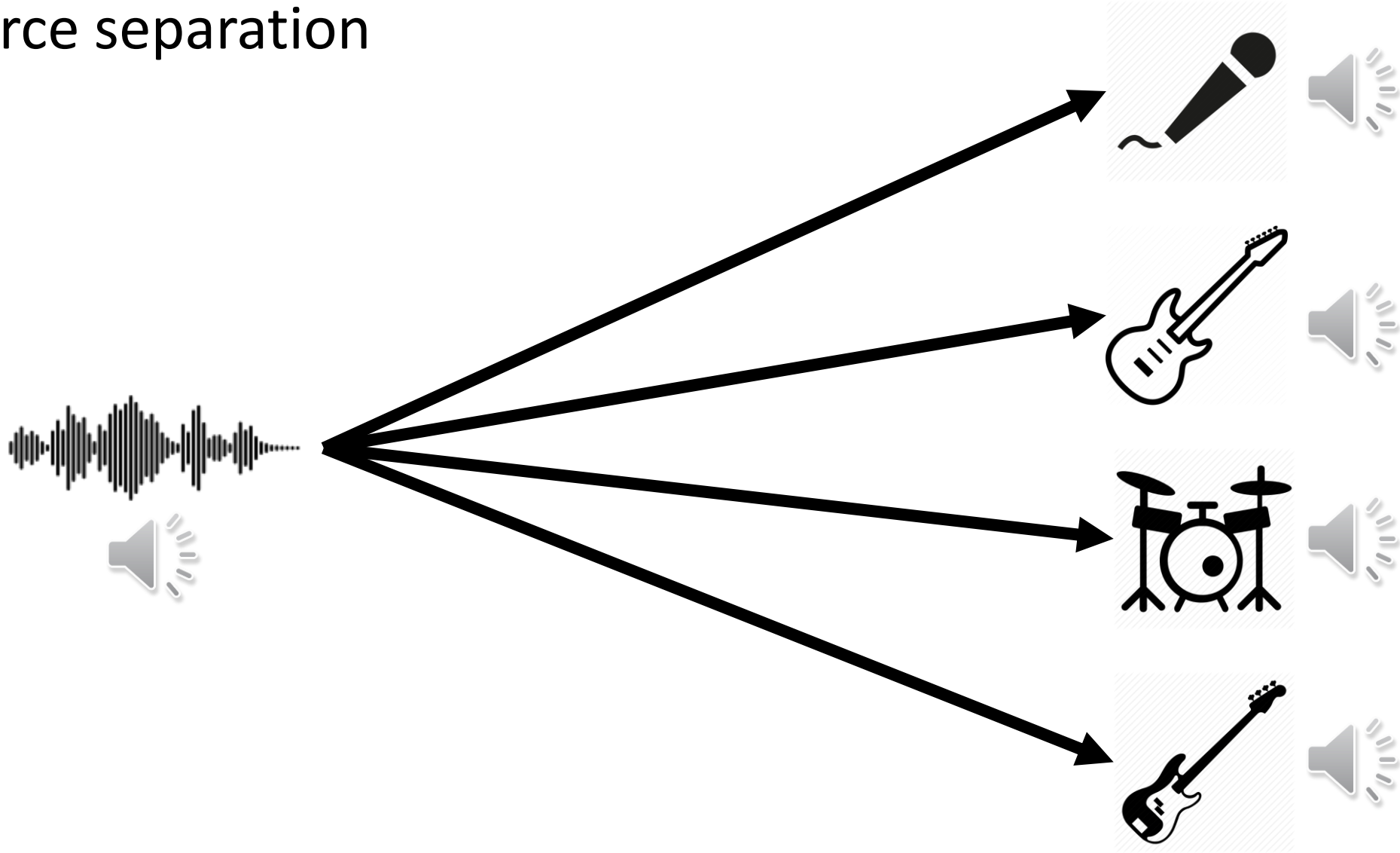




Outline

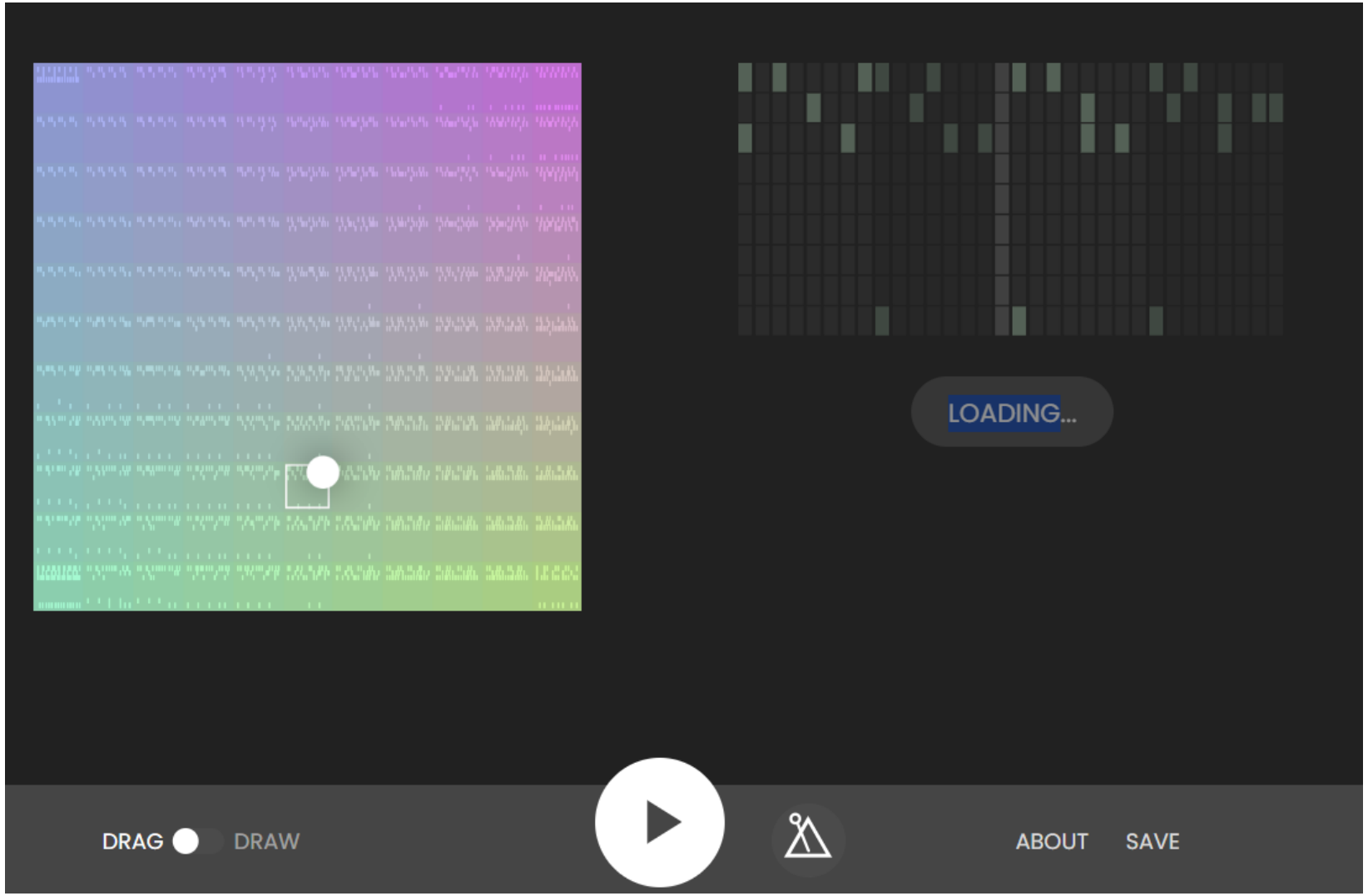
- Music applications
- Artificial intelligence
- Machine learning
- Neural networks
- Deep learning
- How deep learning works
- Suggestions for newbies

Source separation



https://github.com/deezer/spleeter

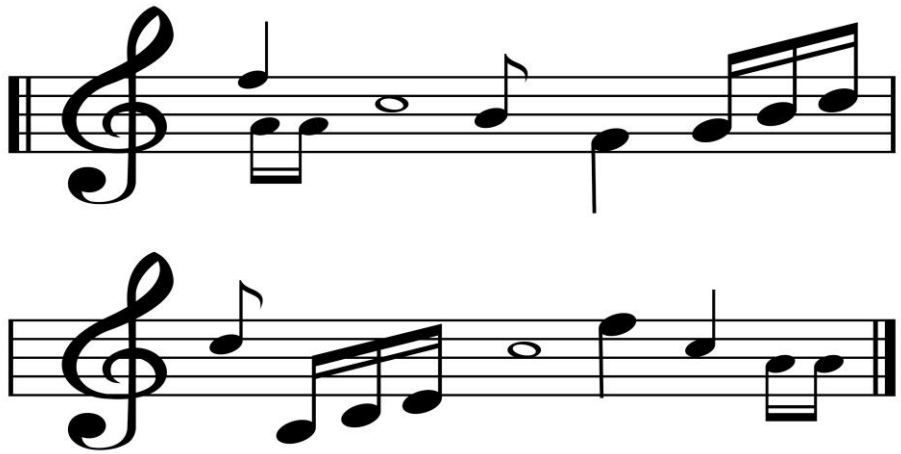
Beat blender



https://experiments.withgoogle.com/ai/beat-blender/view/

Music (symbolic) generation based on videos



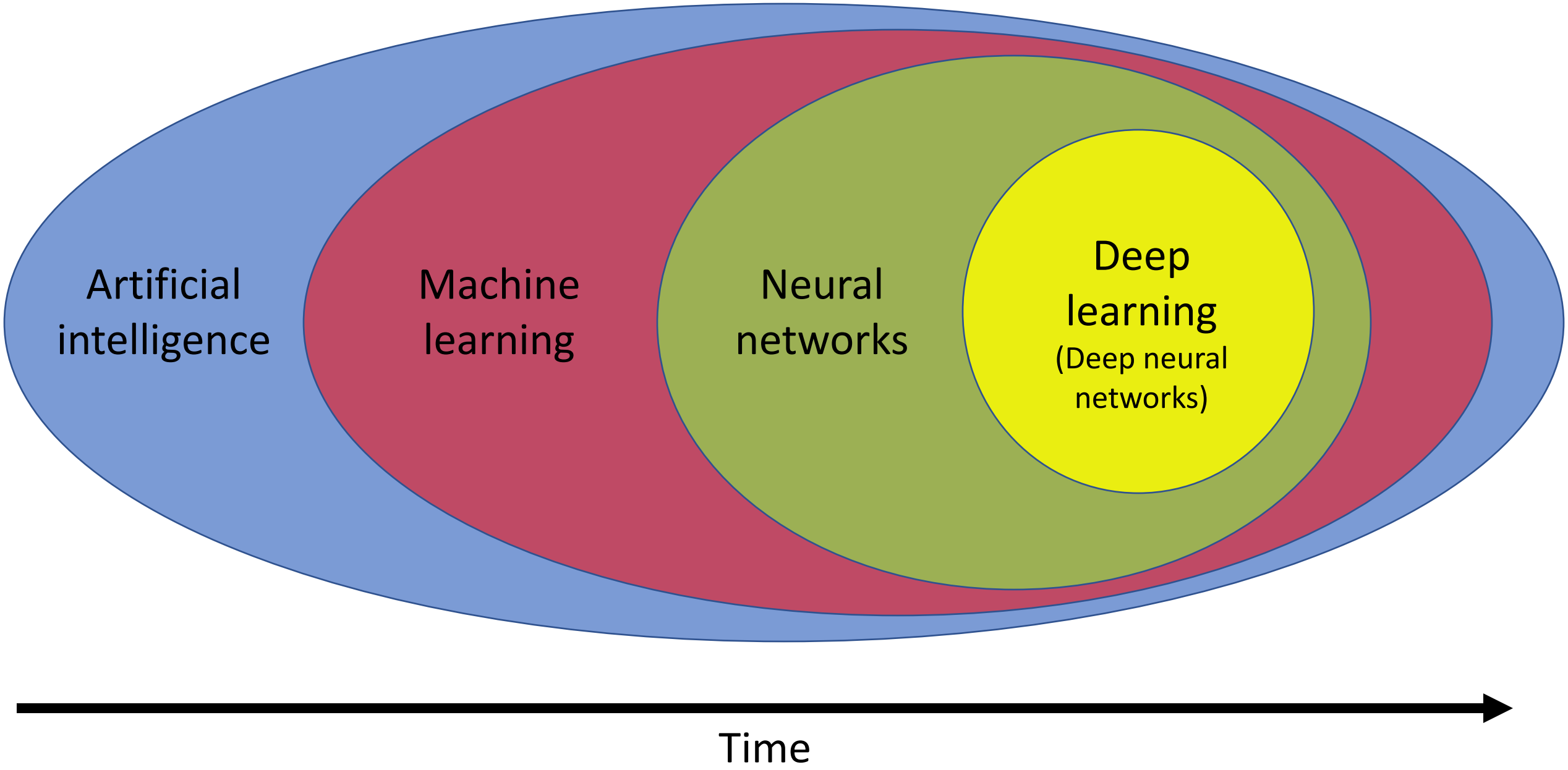




Emotion prediction:
80% Sadness
14% Joy
5% Fear
1% Surprise
0% Disgust
0% Anger

https://serkansulun.com/emsync

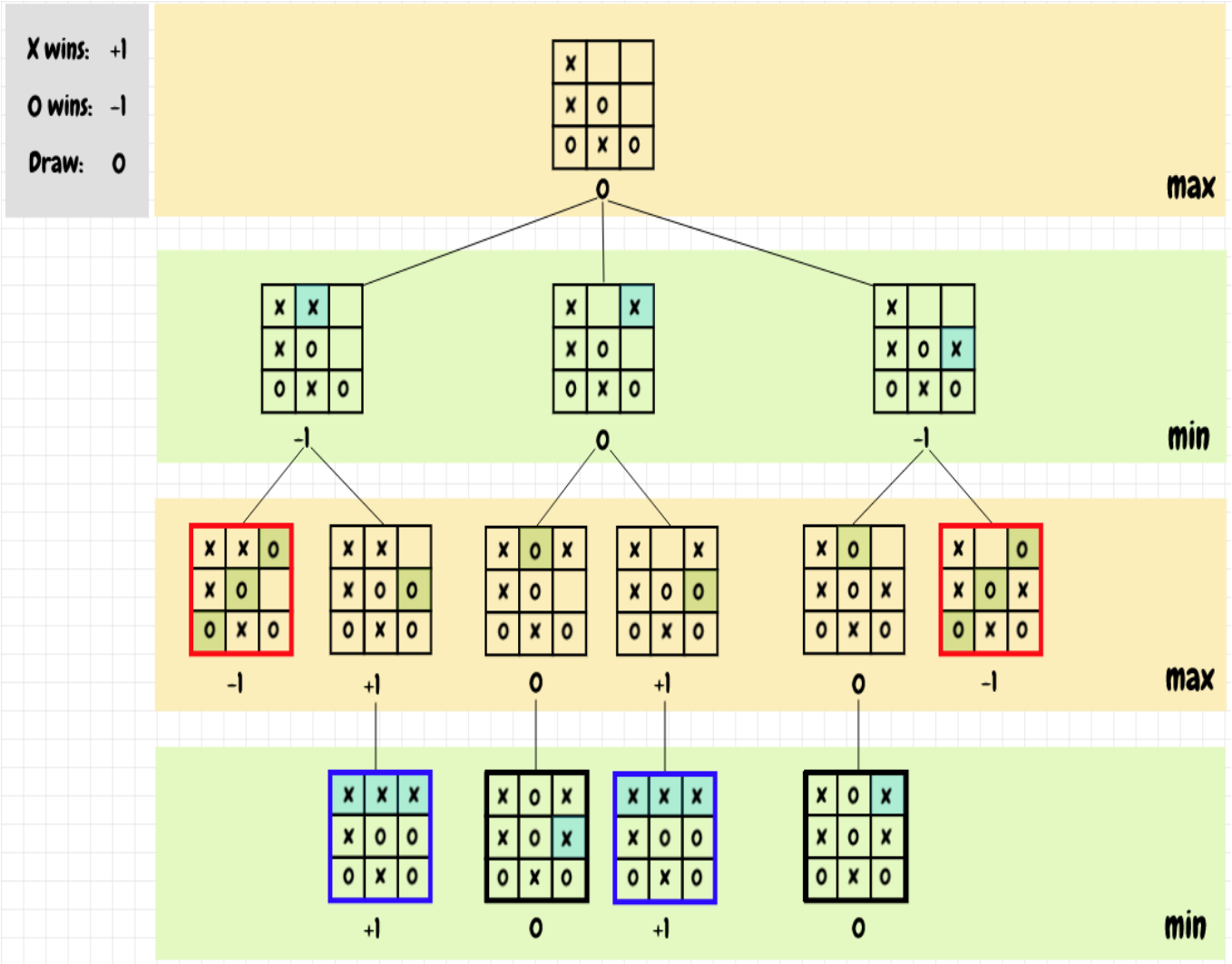
So how does this work?


Artificial intelligence
Machine learning
Neural networks
Deep learning
(Deep neural networks)
Time

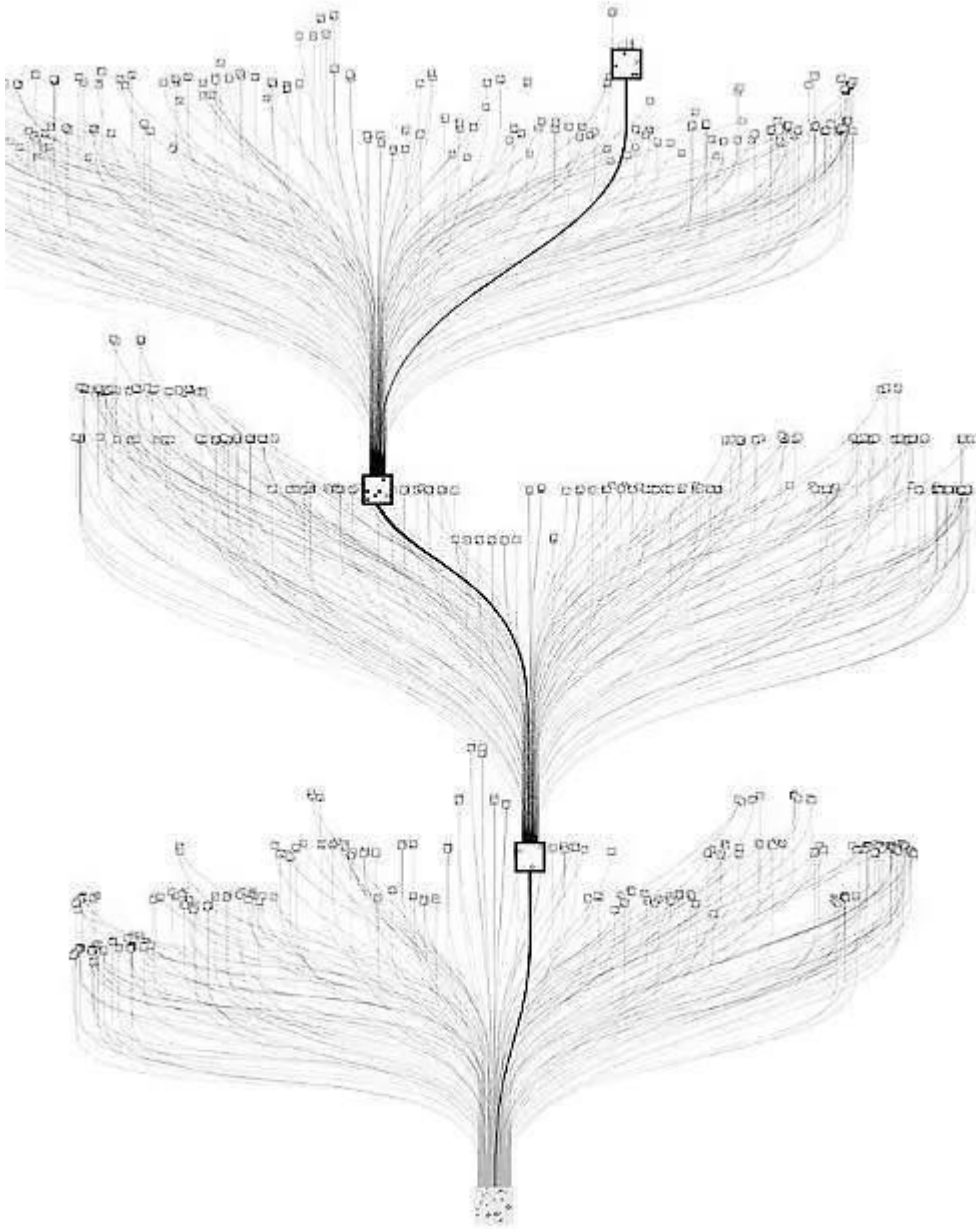
Artificial intelligence: Intelligence demonstrated by machines

Machine learning: Computer algorithms that improve automatically through experience and use of data

Artificial intelligence without machine learning



Rules of the game and the playing strategy is coded by humans.





Artificial intelligence without machine learning



Template





Traditional programming

Programs the rules directly.

```
if template_match_score > threshold:
   donut
else:
   not donut
```

Machine learning programming

Programs various methods to learn the rules from data.

```
for sample in dataset:
   try to classify region
   change to reduce error
   (repeat)
```

Machine learning (classification)

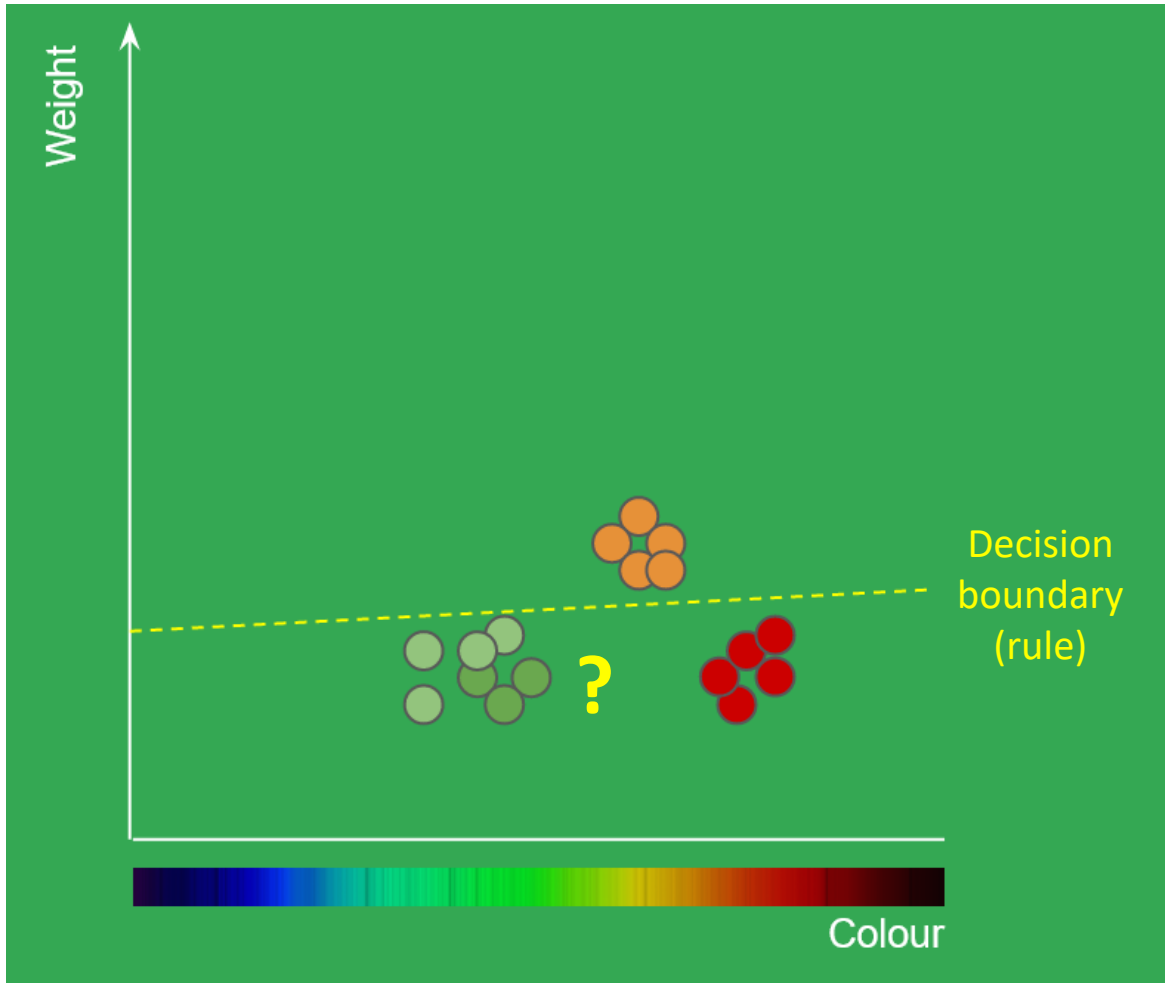


or



Dataset has 5 oranges and 12 apples:

Label	Colour	Weight
Apple	0.4	152
Orange	0.6	181
Apple	0.8	148
…	…	…



Weight and colour are features.
There can be many features.
Label is also called target or ground-truth.

Machine learning (classification)

Model can also predict the probability of being an apple.

0.93 → Apple 0.12 → Orange

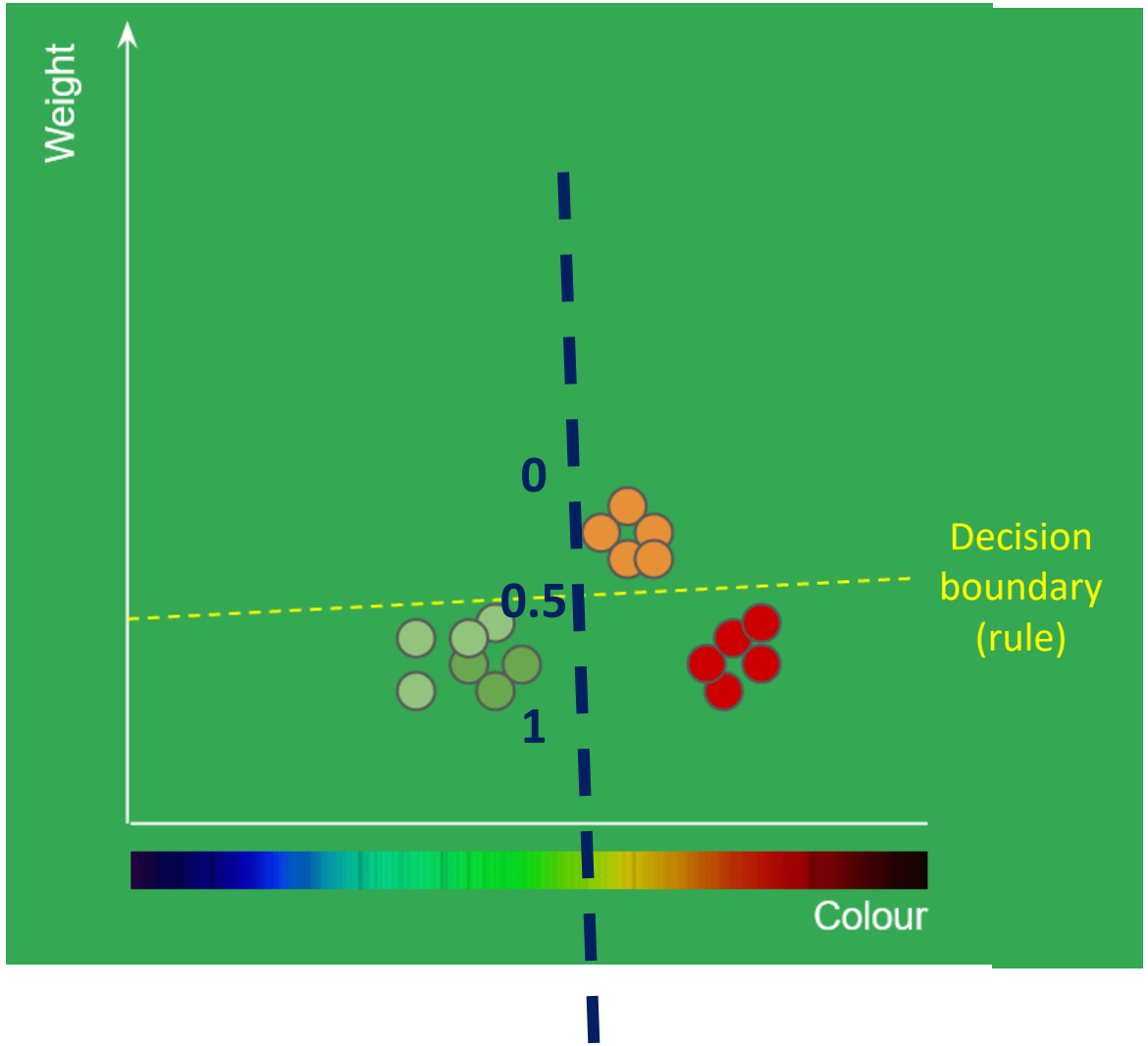
Some models can predict the probability of both classes simultaneously.

Apple: 0.63 , Orange: 0.37 → Apple

This useful in multi-label classification:

Apple: 0.11 , Orange: 0.18, Pear: 0.71 → Pear

ImageNet object classification challange has 1000 labels, and state-of-the-art models have 90% accuracy.



Project any data point onto this line, to find probability of being an apple, and threshold at 0.5 to make final decision

Machine learning (regression)

Output is a number, not a label.

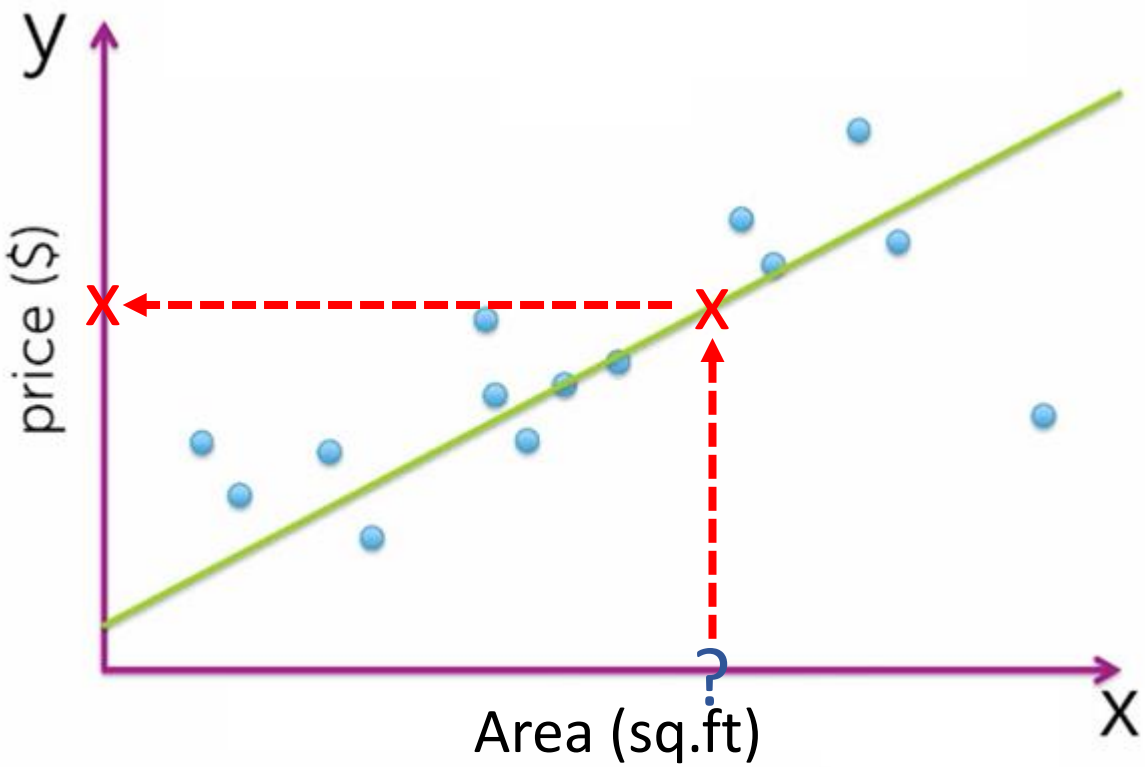
Dataset has 14 houses:

Price	Area (sq.ft.)
230,000	1712
420,000	2356
…	…

Area is the feature.
Price is target or ground-truth.

Predict house prices





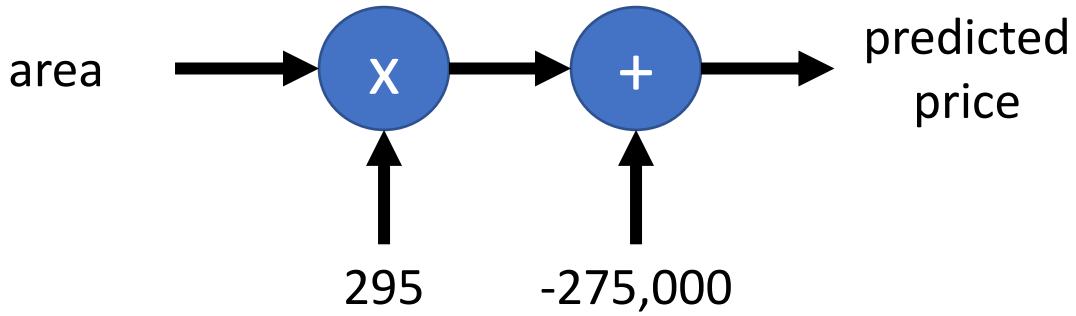
$$predicted\ price = 295\ area\ - 275{,}000$$

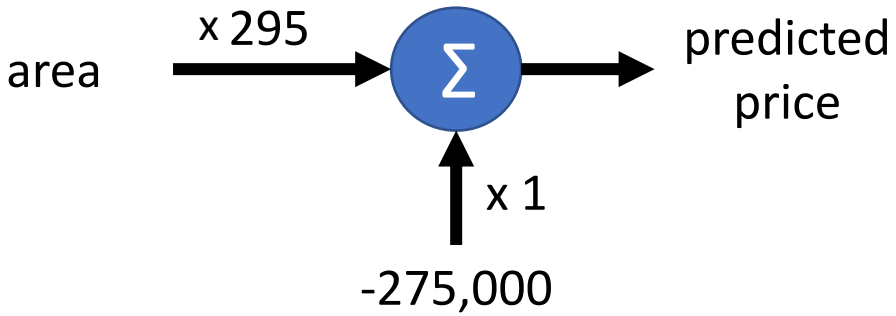
A different representation using neural networks

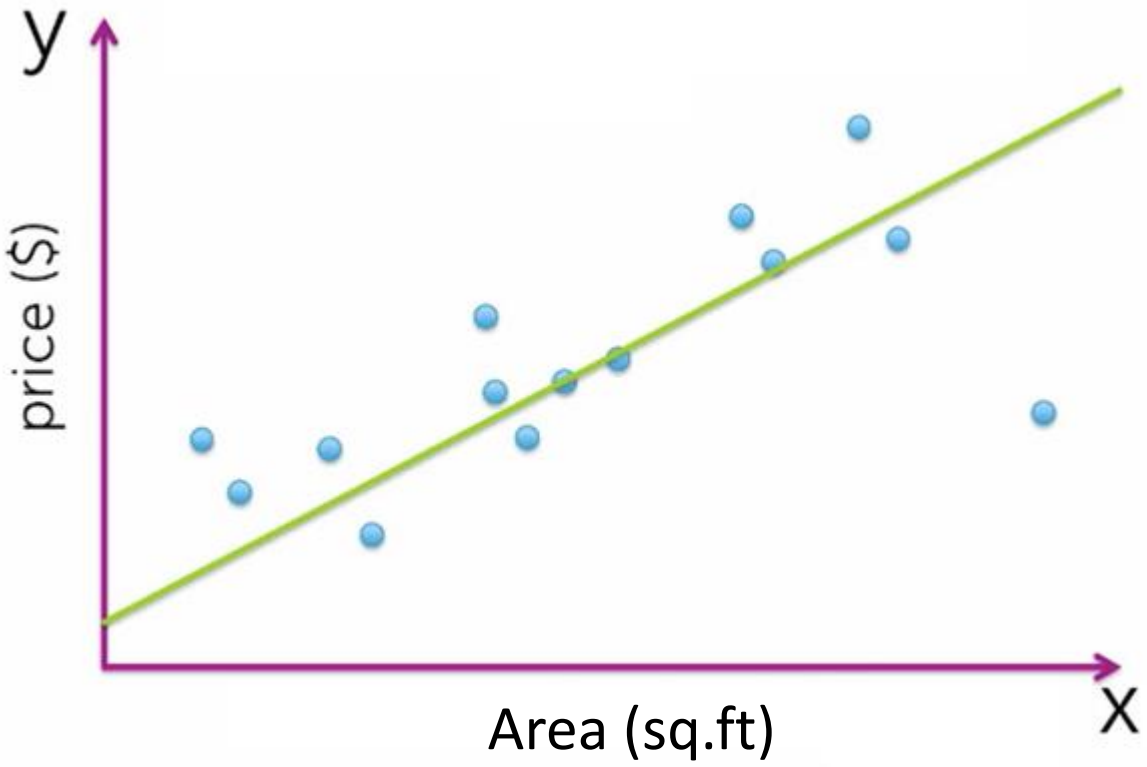
$$predicted\ price = 295\ area\ - 275{,}000$$

$$\hat{y} = wx + b$$

weight bias



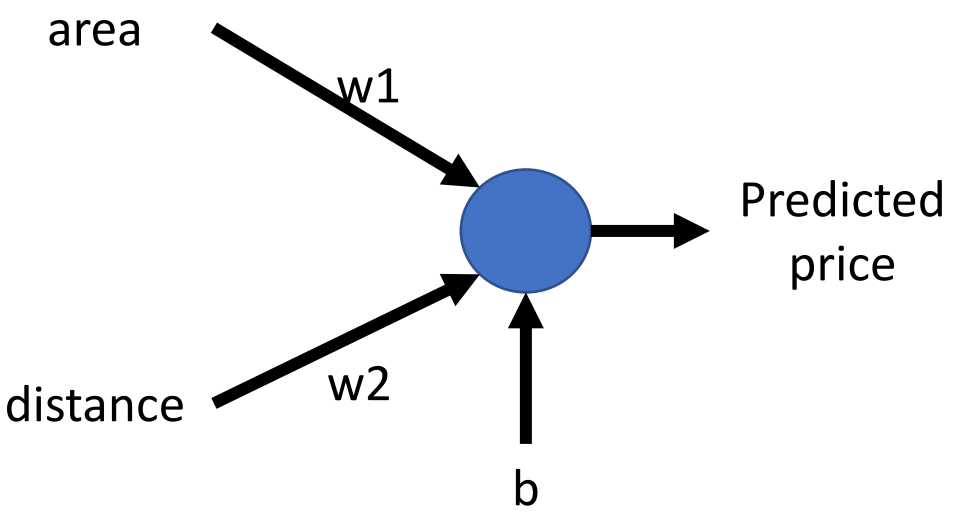


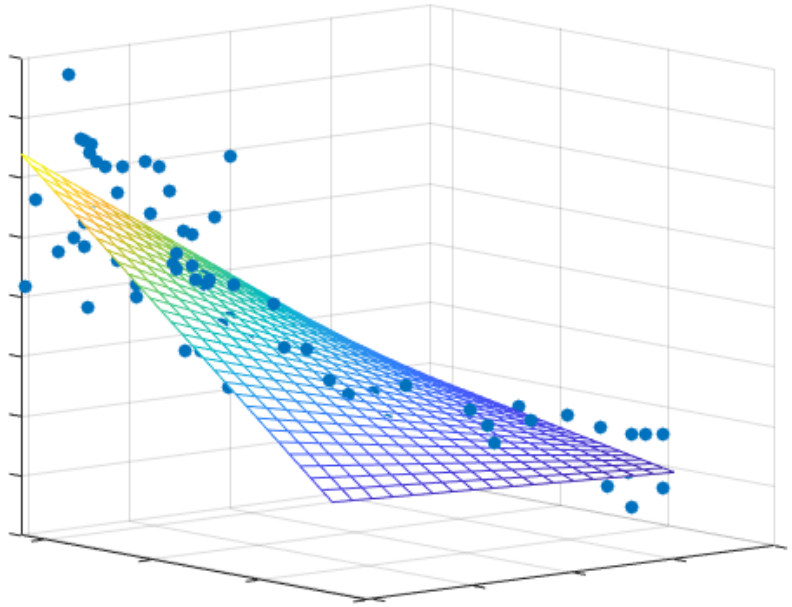


What if we had two features?

What if we had two features: area and distance to city center

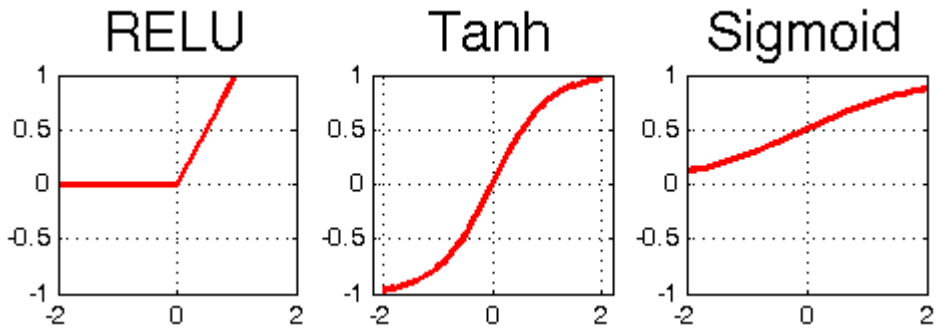
$$predicted\ price = w_1\ area + w_2\ distance + b$$



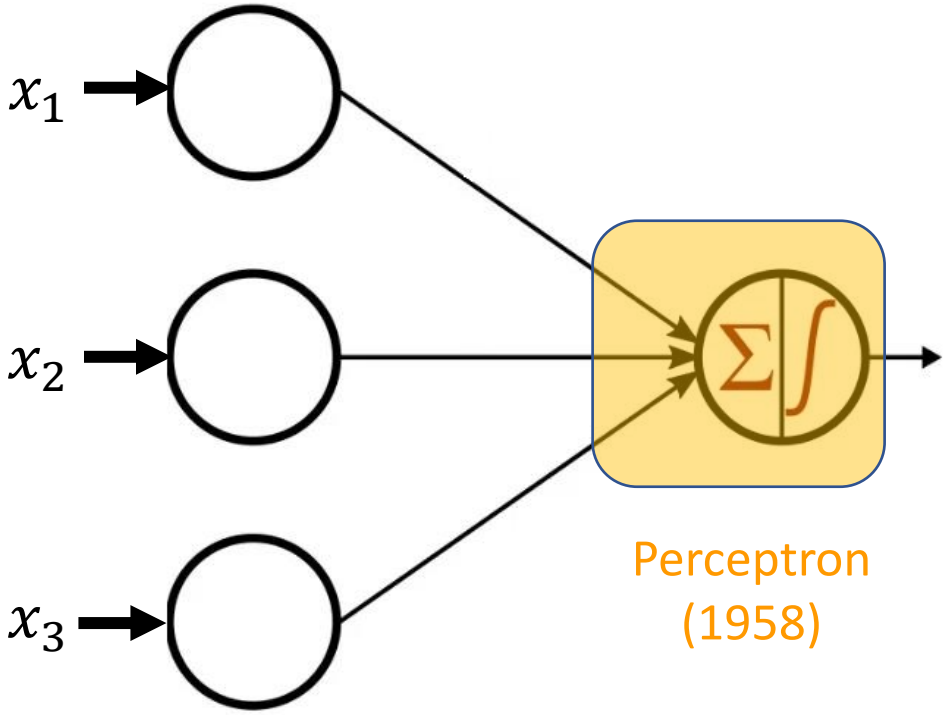


What if we had a nonlinear function?

Use nonlinear function <u>inside</u> the network

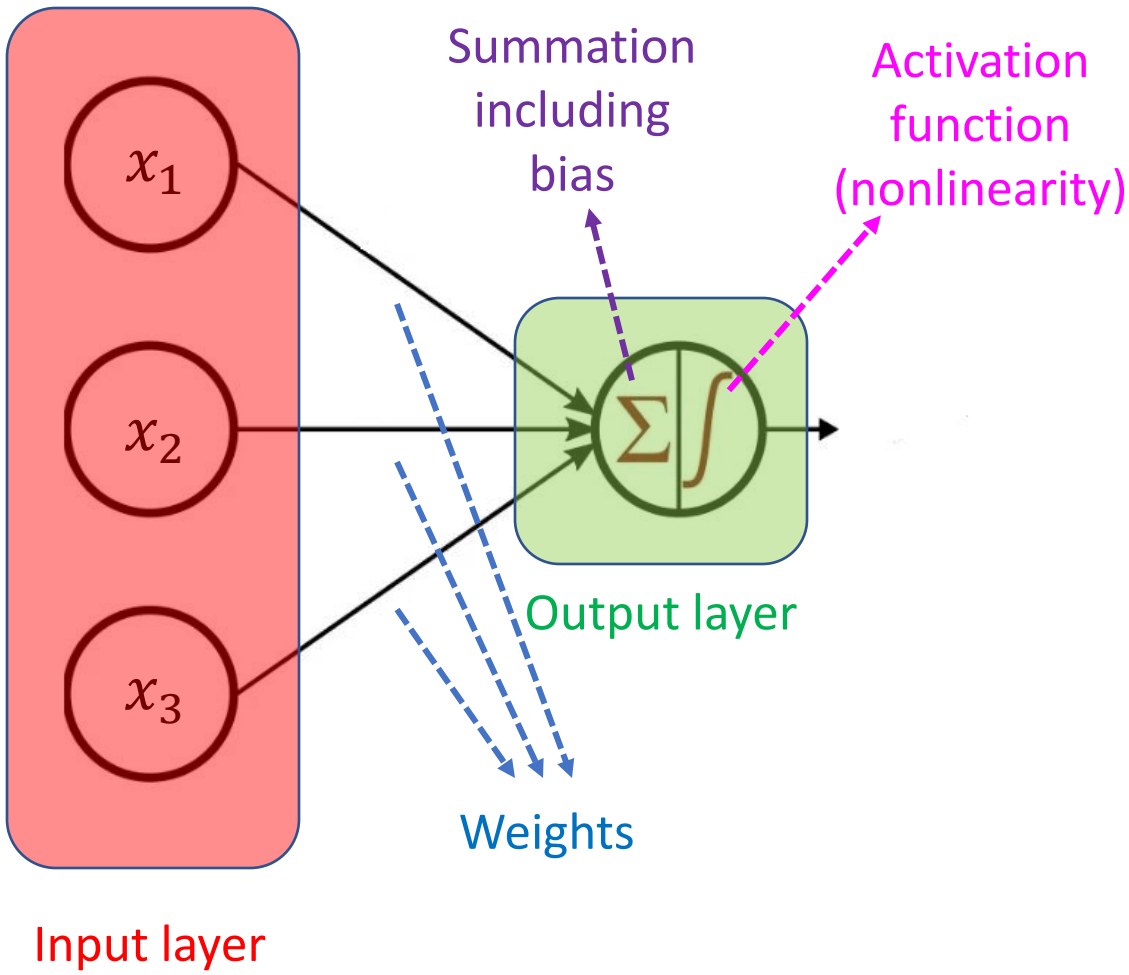


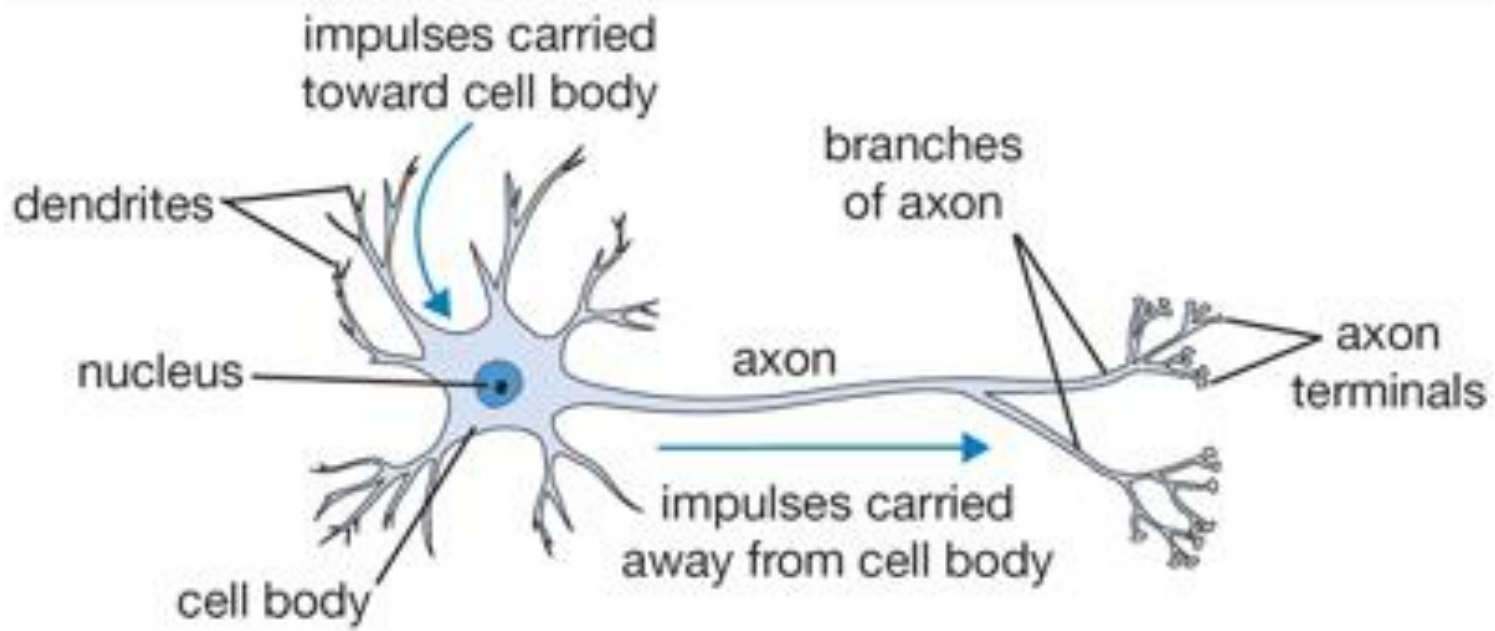
Representation and vocabulary

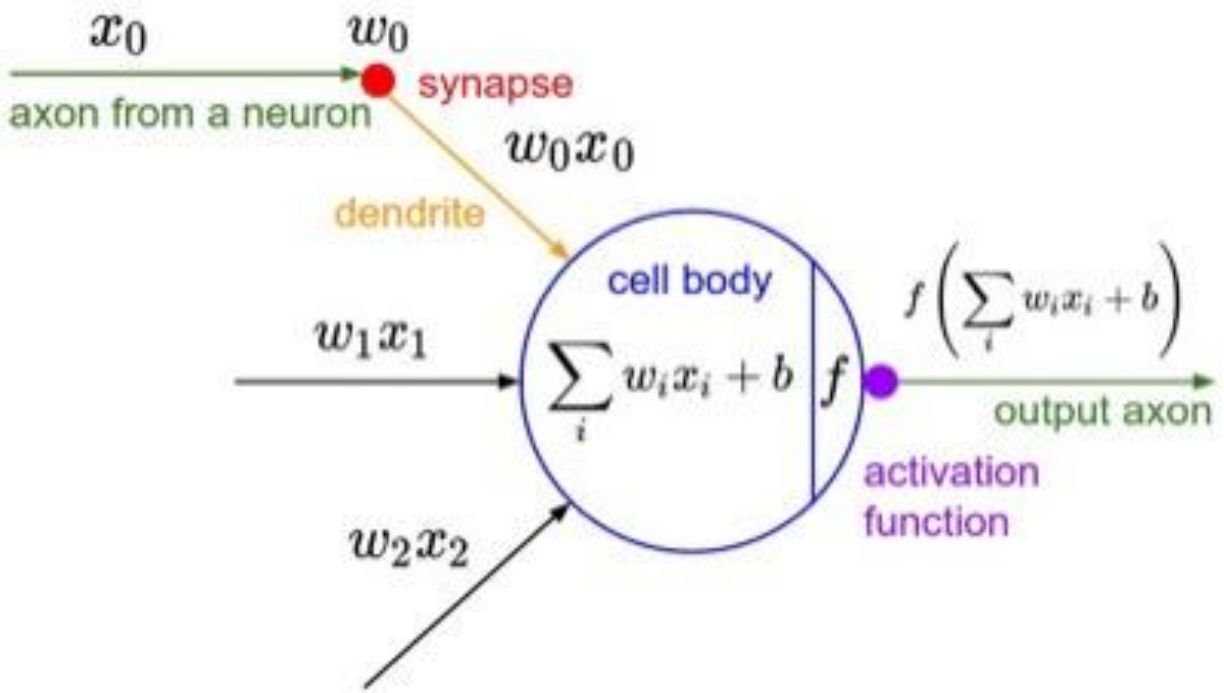


Perceptron
(1958)

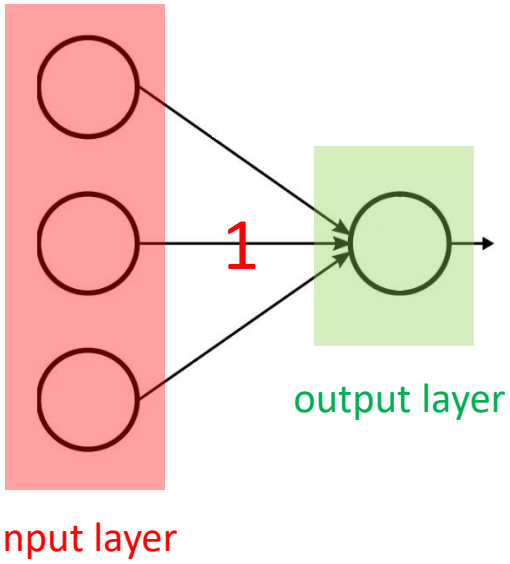
Representation and vocabulary



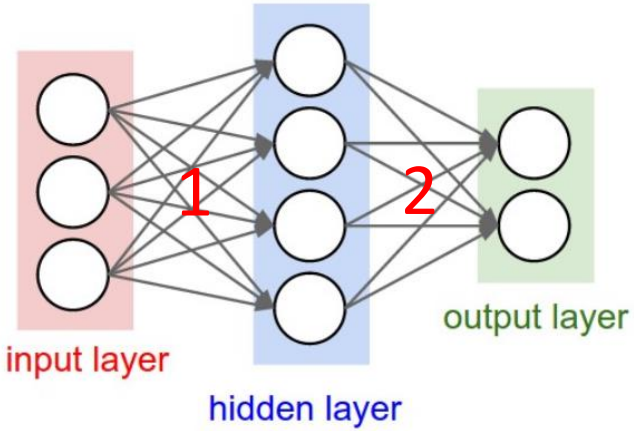

impulses carried
toward cell body
dendrites
branches
of axon
nucleus
axon
axon
terminals
impulses carried
away from cell body
cell body


x_0
w_0
synapse
axon from a neuron
$w_0 x_0$
dendrite
cell body
$w_1 x_1$
$\sum_i w_i x_i + b$
f
$f\left(\sum_i w_i x_i + b\right)$
output axon
activation
function
$w_2 x_2$

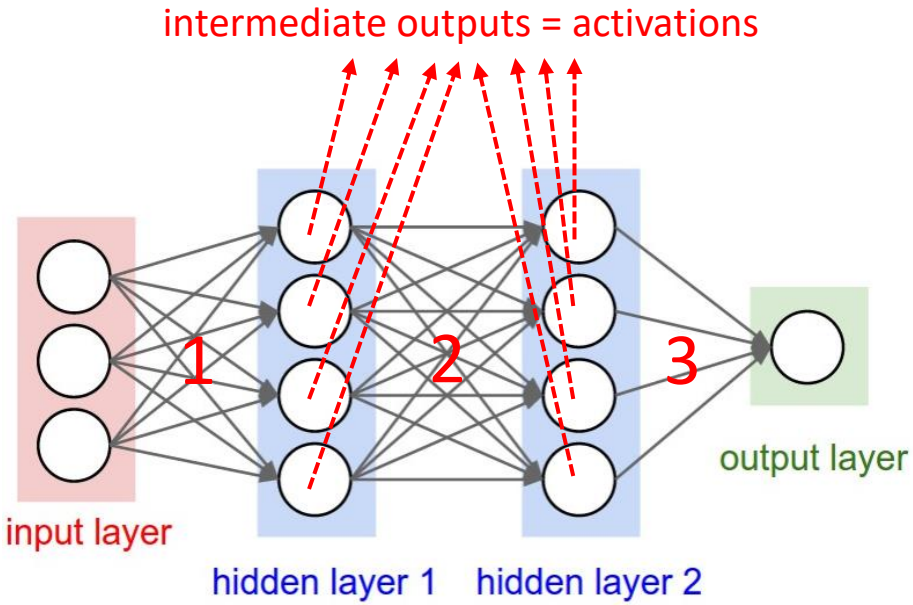
Representation and vocabulary



1-layer neural network with
3 inputs and 1 output

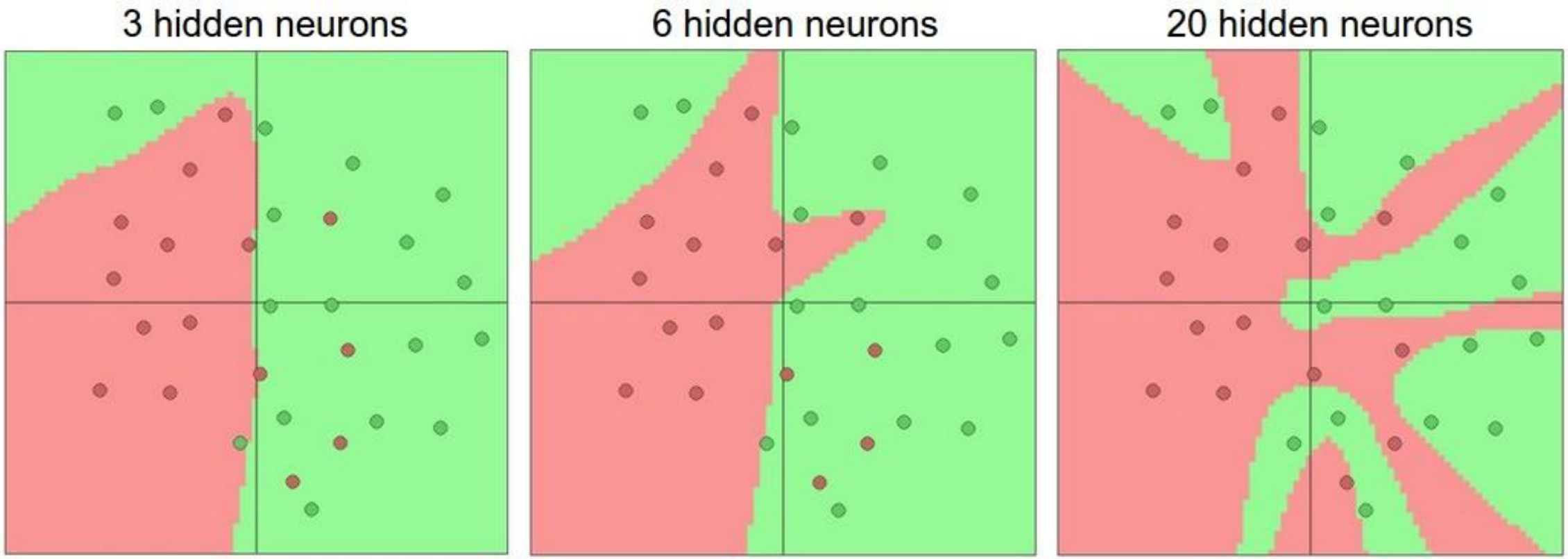


2-layer neural network with
3 inputs, 1 hidden layer
with 4 units (neurons),
and 2 outputs



3-layer neural network with
3 inputs, 2 hidden layers
each with 4 units (neurons),
and 1 output

Representation power



Features



or



Our features were: color and weight

Are these enough for real-world application?





How about some other features: shape, surface smoothness, color consistency, stem length/color, ...

How many useful features are there?

How to extract these features?

Can these features be used to classify other fruits? If not, the program has to be re-written.

How about animals?

Raw pixels as features

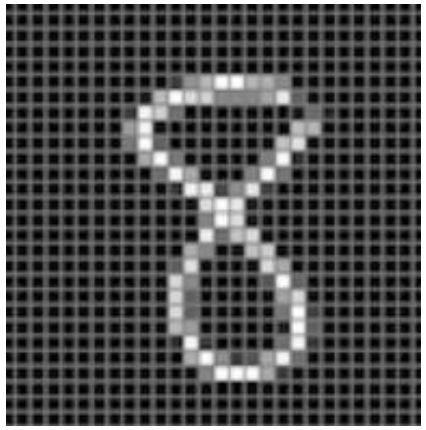
(Each pixel is a number from 0 to 255. Pure black is 0, pure white is 255.)

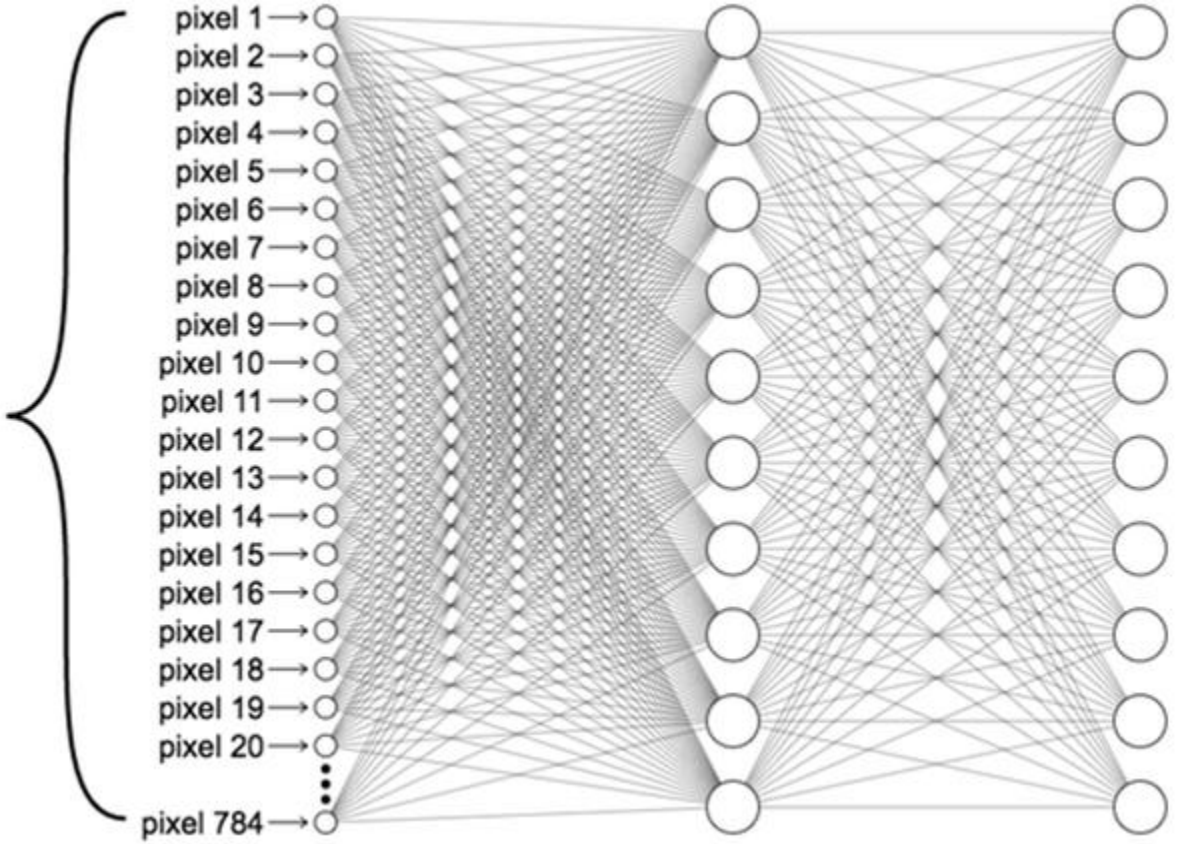
As humans, we know that useful features are implicitly present in the raw image:

- number of holes
- width-to-height ratio
- crossings points

…

What if the neural network could extract any useful feature by itself?

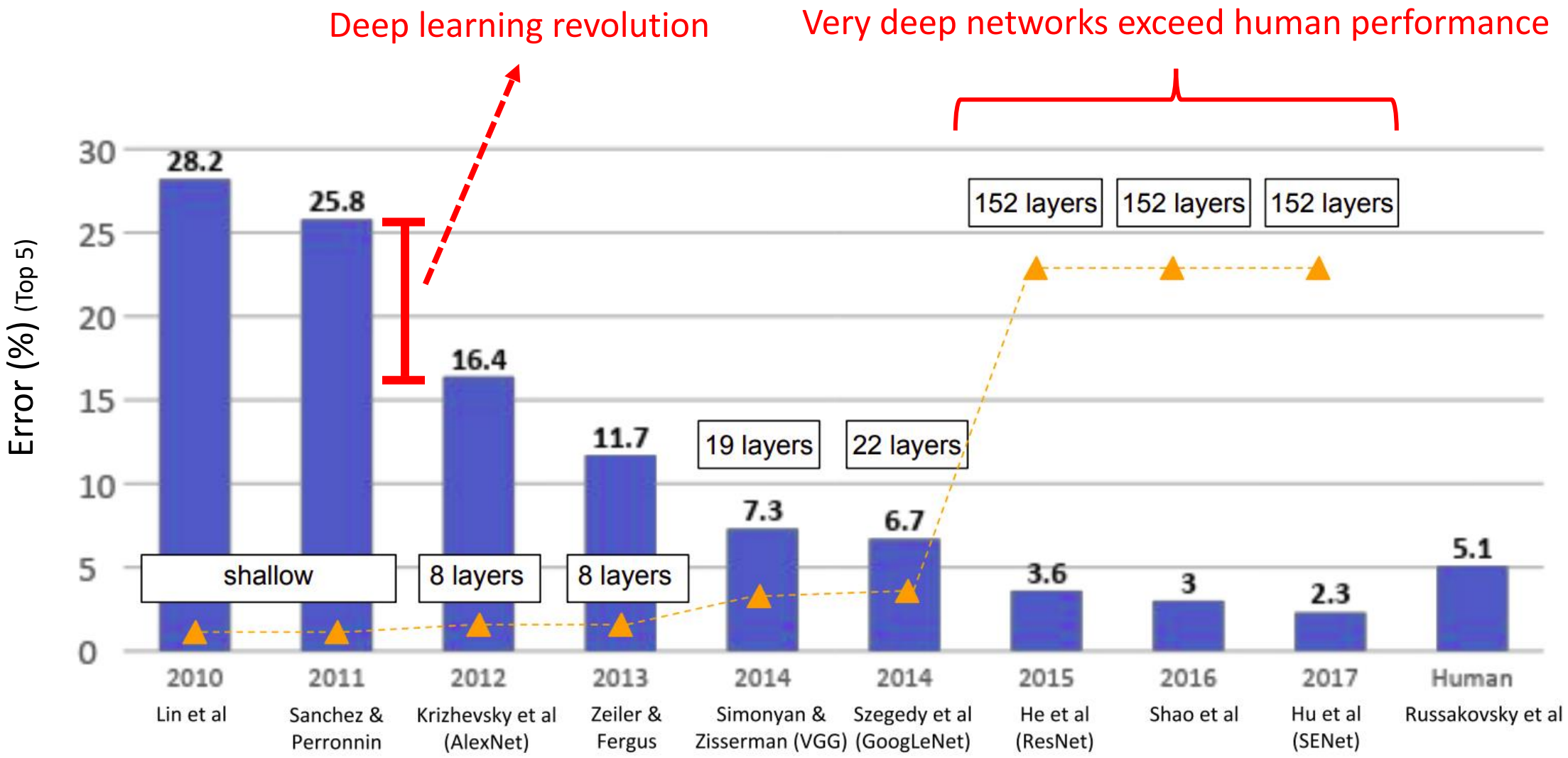




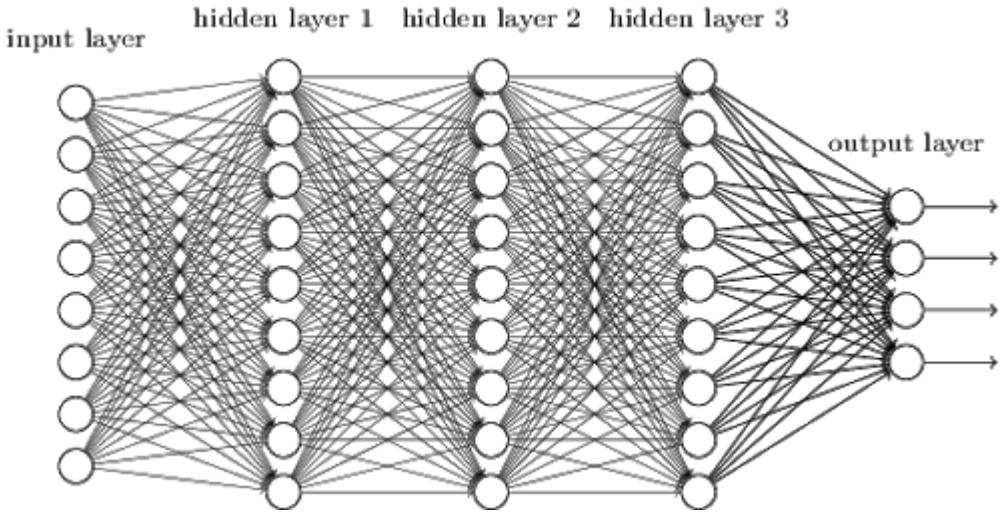
Solution: Deep neural networks

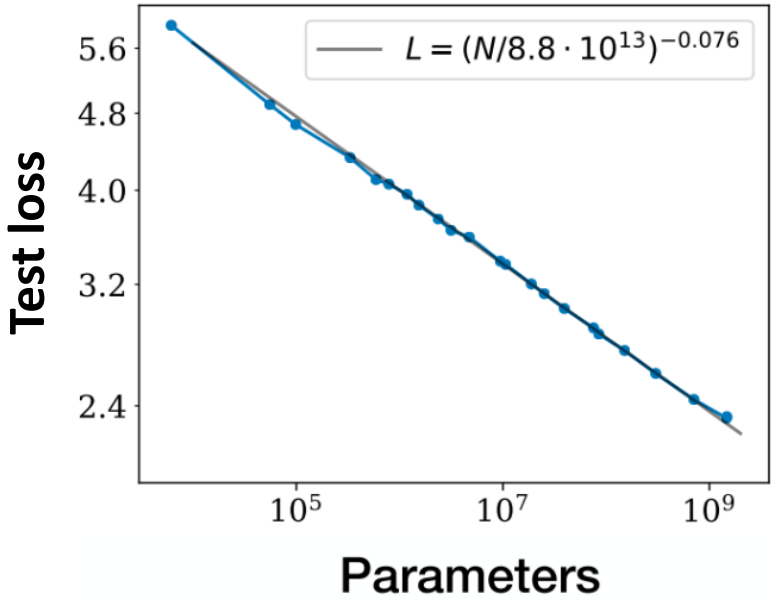
Effectiveness of deeper neural networks

Image classification challange results over years



Deep neural networks (DNNs)







2015 ImageNet winner ResNet: 60 million weights (152 layers)
2020 OpenAI LLM GPT-3: 175 billion weights
2025 Alibaba MLLM Qwen3-Max: 1 trillion weights

larger models + larger datasets → higher performance

How do neural networks learn? (How are neural networks trained?)



DNN



input

target

Dataset







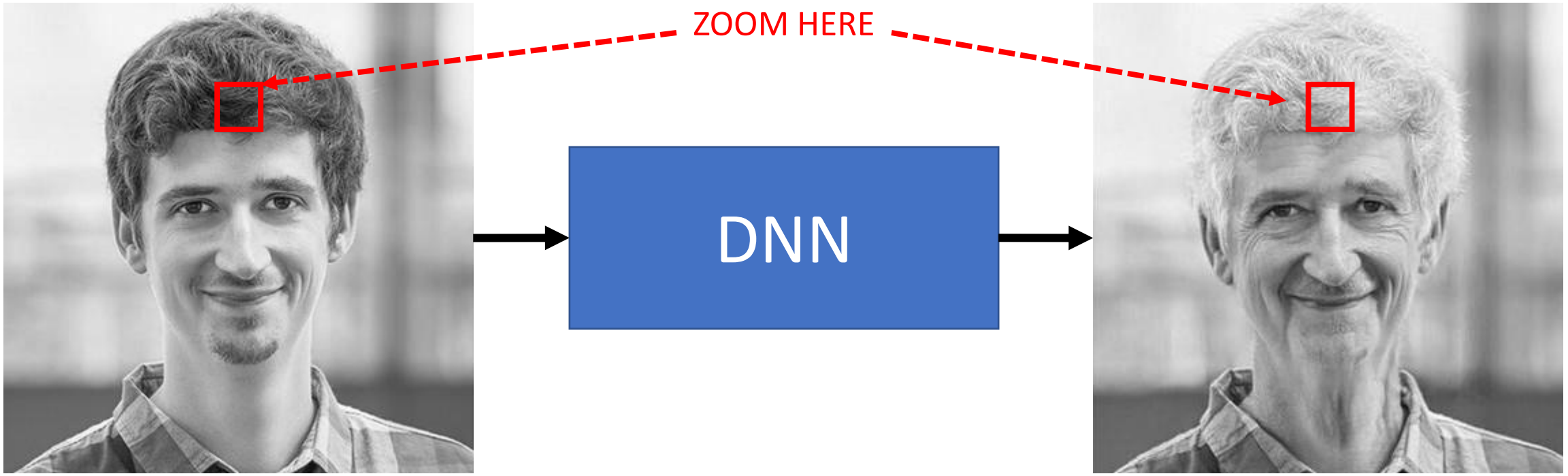






and **millions** more…

How do neural networks learn? (How are neural networks trained?)



input

target

Given a 400x400 image, both input and target have 400x400=160,000 units (numbers)

(Think of it as 160,000 features and 160,000 labels)

How do neural networks learn? (How are neural networks trained?)



input
(patch of black hair)

target
(patch of gray hair)

13	11	8
14	12	9
12	10	7

multiplication,
addition,
nonlinearity

144	152	155
139	142	145
136	140	143

How do neural networks learn? (How are neural networks trained?)

At first, all DNN weights are initialized randomly.



NON-TRAINED
DNN



input

output

How do neural networks learn? (How are neural networks trained?)

Every training has an objective: **Minimize error** (loss)

error = | target - output |



=

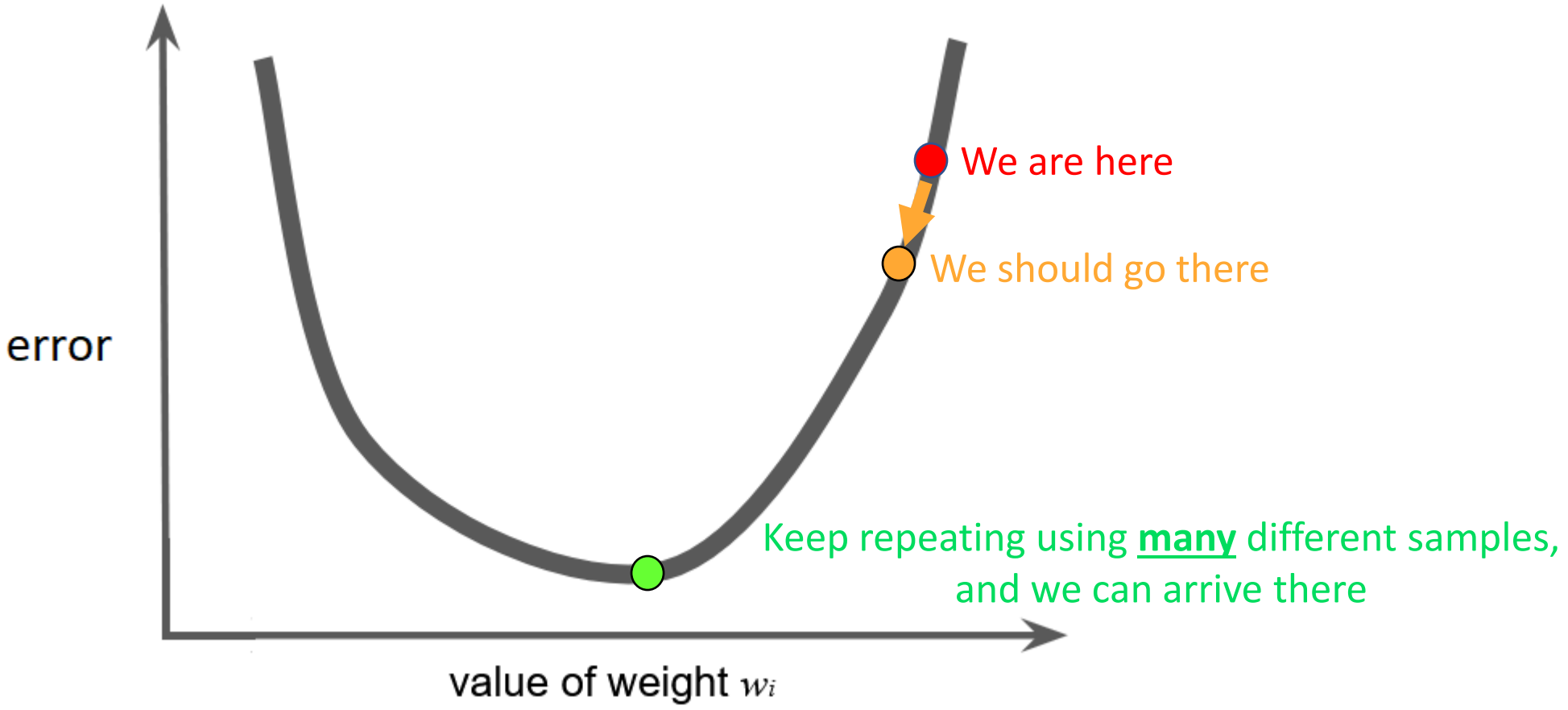


-



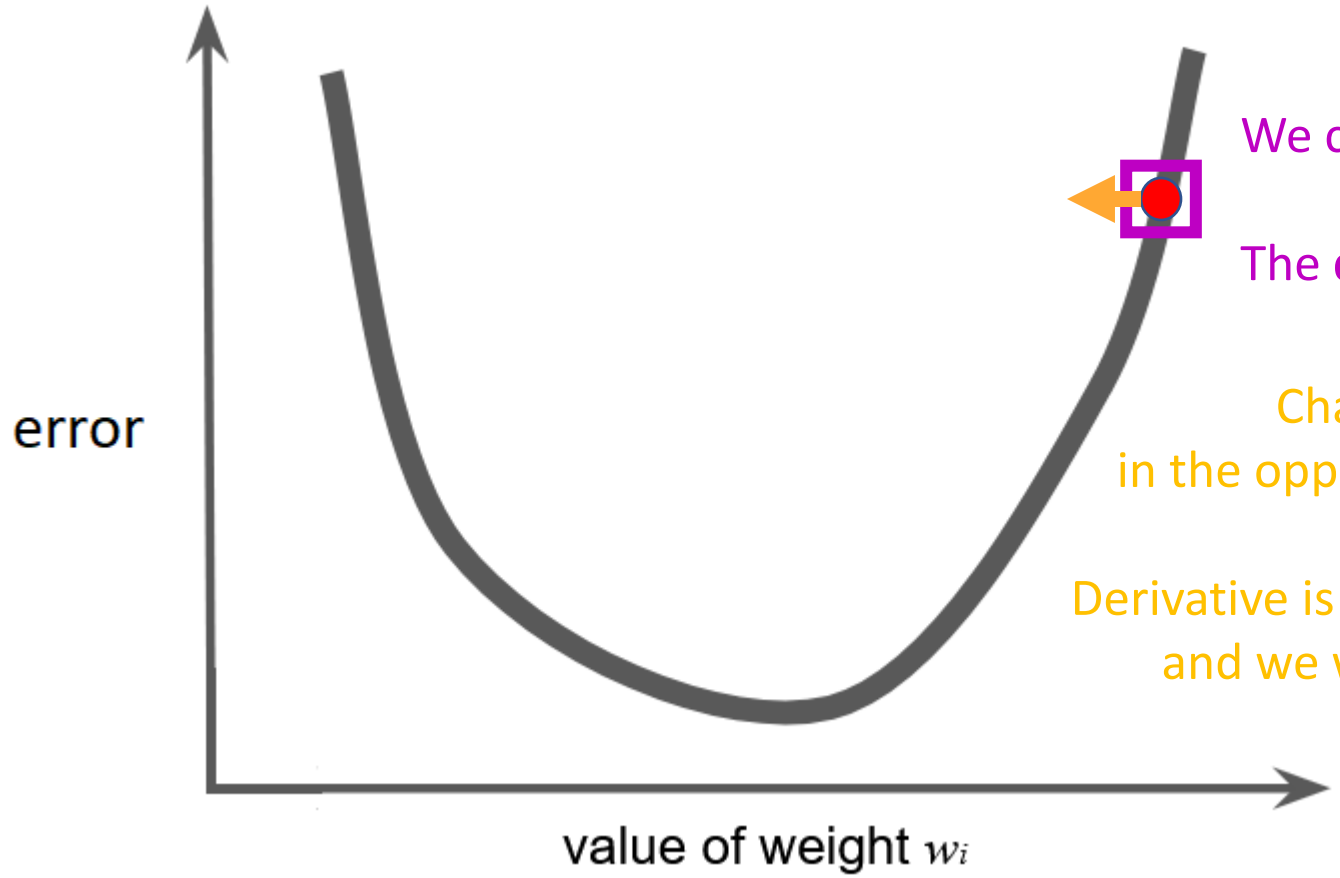
How do neural networks learn? (How are neural networks trained?)

Change the weights slightly, so that the error is reduced.



How do neural networks learn? (How are neural networks trained?)

In reality, we cannot see this entire graph.
(We'd need to compute output for infinitely many values)



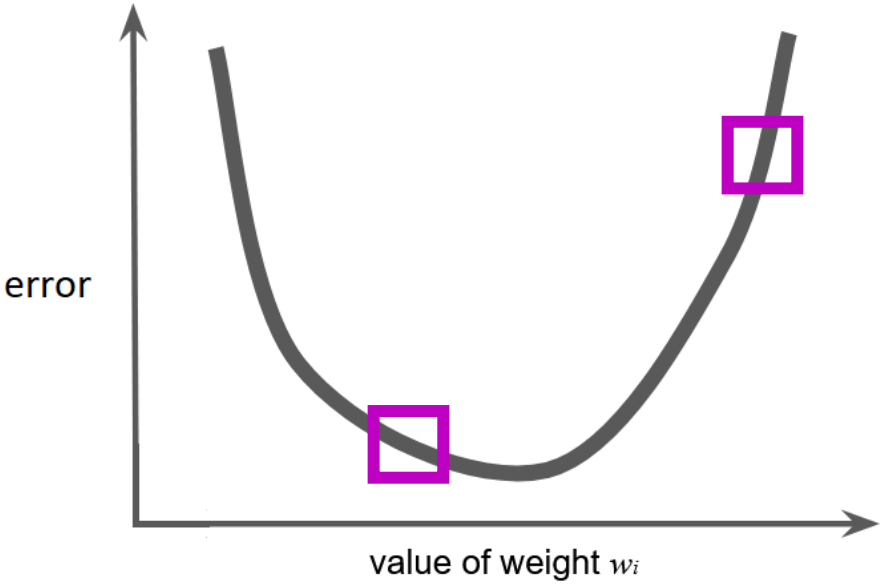
We can only see this much:

The **derivative** at that point

Change the weight slightly
in the opposite direction of the derivative

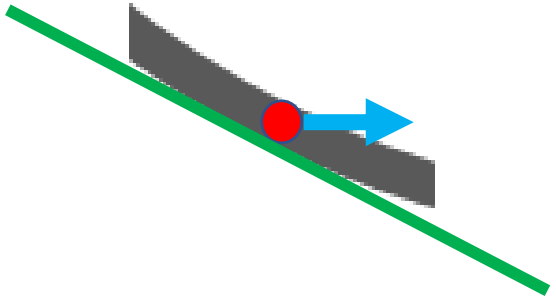
Derivative is positive for increasing functions,
and we want to minimize (decrease).

How do neural networks learn? (How are neural networks trained?)



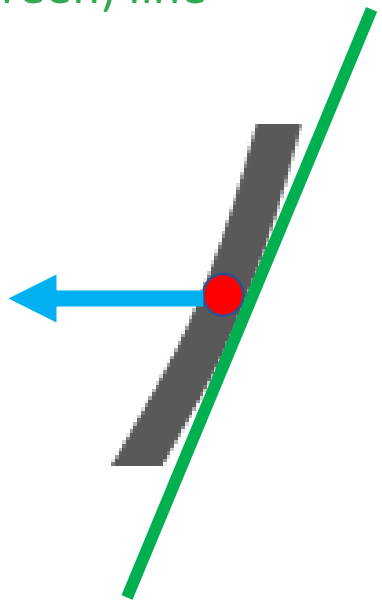
Let's zoom in

derivative = slope of tangent (green) line



Derivative is negative
and has small magnitude

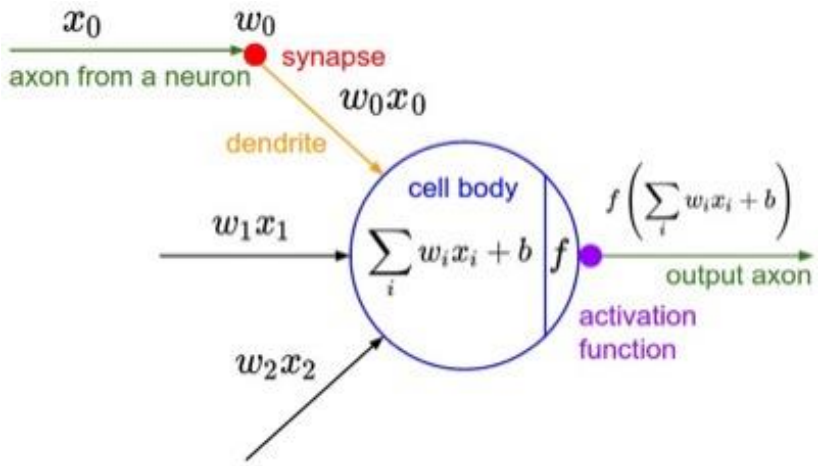
Take a small step to the right

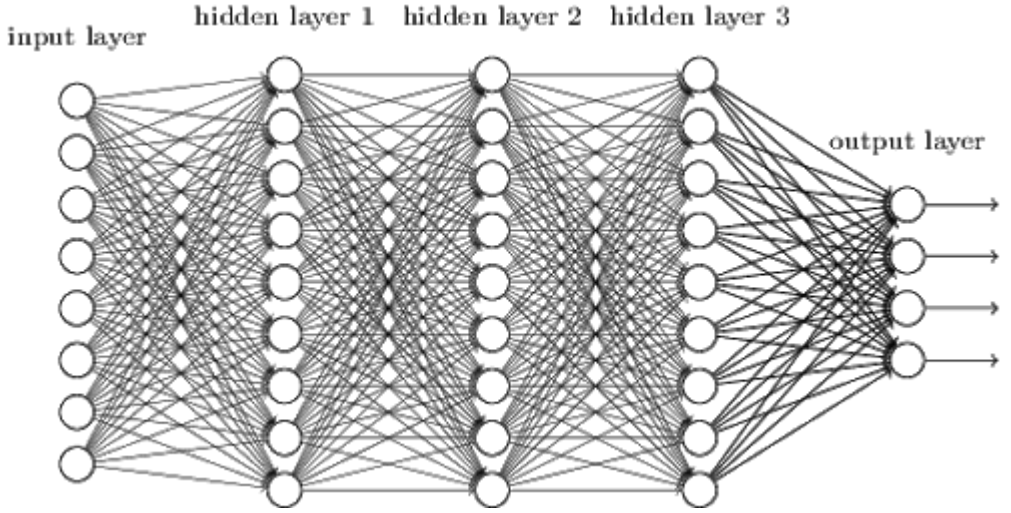


Derivative is positive
and has large magnitude

Take a large step to the left

How to calculate the derivatives?





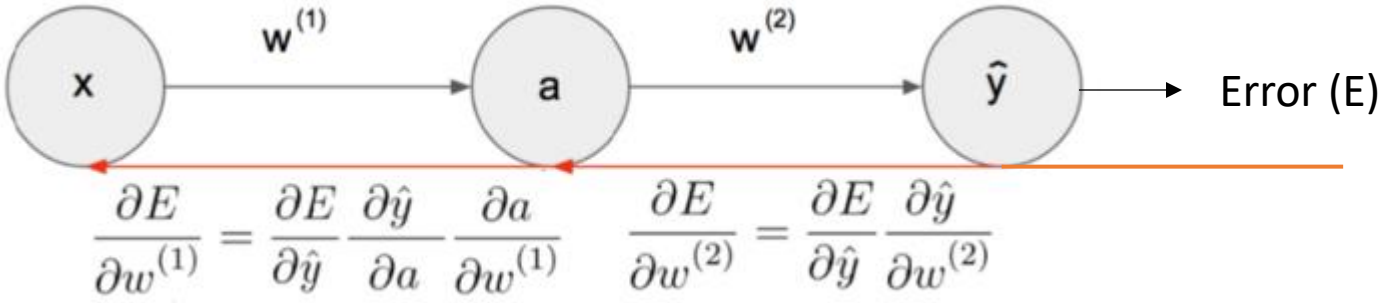
But there are many of these operations and they add up to a highly complex function:

The neural network consists of elementary operations such as:

- Multiplication
- Addition
- Nonlinear function

The derivatives of these operations are known and implementable.

So use **chain rule**



How to calculate the derivatives?

Doing this is for a computer program is called **automatic differentiation.**

It is non-trivial, there are large textbooks on this topic.

The field of DNN is also evolving, new architectures and operations become standard practice.

So we use deep learning libraries:





They perform automatic differentiation with one line of code, and they are kept up-to-date.

They are free and open-source.

They have extensive documentations, tutorials and examples.

They have large communities where you can get answers for your questions.

Pytorch code for Faceapp example

```python
import torch
import torch.nn as nn
from torchvision.io import read_image
import os

class NeuralNet(nn.Module):
    def __init__(self):
        super(NeuralNet, self).__init__()
        self.model = nn.Sequential(
            nn.Conv2d(1, 16, 5),
            nn.ReLU(),
            nn.Conv2d(16, 32, 5),
            nn.ReLU(),
            nn.Conv2d(32, 1, 5))

    def forward(self, input):
        output = self.model(input)
        return output

input_images = os.listdir("path/to/input_images")
target_images = os.listdir("path/to/target_images")
n_samples = len(input_images)

model = NeuralNet()
optimizer = torch.optim.SGD(model.parameters())

for i in range(n_samples):
    input = read_image(input_images[i])
    target = read_image(target_images[i])
    output = model(input)
    error = (target - output) ** 2
    optimizer.zero_grad()
    error.backward()
    optimizer.step()
```

Multiplication

Nonlinearity

3 layer neural network

Read dataset file paths

Initialize network

Trains network

Repeat for entire dataset

Read input and target images

Run network

Calculate error

Reset derivatives from previous run

Calculate derivative

Update model

Practical consideration: GPU

To speed things up, or to train larger models, using a GPU is essential.

CPUs are 10 to 100 times slower.

Cloud computing services do offer GPUs:

- Google Colab (free with time limit)
- Google Cloud
- Amazon Web Services (AWS)
- Microsoft Azure

Search for student credits, free trials, new services with introductory offers.

Further recommendations

Learn **Python**, it is one of the easiest, most popular and most useful languages.

If you know any other language, you can start with Python immediately and learn while using it.

Pytorch is much simpler than Tensorflow.

There are tons of examplary codes on **Github**.

Familiarize yourself with **Linux**, deep learning tools are unstable on Windows.

Ubuntu is a very popular one, it is free, you can install and use it alongside Windows (dual-booting).

Visual Studio Code is a lightweight IDE, and it enables remote debugging for cloud computing.

Github Pro (AI) is free for students. To learn, it's better "ask" AI rather than letting it do all the work.


Questions ?